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Mainstreaming regenerative dynamics for sustainability

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Notions of regeneration have entered discourses in several fields that are relevant for sustainability, including, among others, ecology, agriculture, economics, management, sociology, psychology and chemistry. A review of existing work shows that there are interesting parallels between these fields. By carefully defining key concepts, such as regenerative dynamics, regenerative practices and regenerative momentum, this Review Article offers a generalized framework for understanding regenerative systems. This framework, in turn, promises to spark new insights for sustainability science and practice because it can link hitherto disconnected academic fields and foster new, integrative developments across multiple areas of sustainability practice.

The global sustainability crisis is driven by rapid changes in social and ecological relationships[1], including numerous reinforcing feedback cycles[2,3]. The resulting dynamics are undermining the capacity of the Earth system to reliably provide key functions necessary to sustain human well-being[4]. Whereas degenerative dynamics that compromise a sustainable future have received attention for decades, interest is now growing within multiple domains in the opposite phenomenon: regenerative dynamics that lead to ongoing improvements in the state of a particular system.

In search of solutions to the sustainability crisis, the terms 'regeneration', 'regenerative' and other derivatives thereof have entered sustainability-relevant research and practice in various fields, among others, in the contexts of economic systems[5], business management[6] and production processes flows[7]; agriculture[8] and ecosystem management[9]; as well as human–nature connectedness[10] and psychological and physical well-being[11]. A recent advance in cohering this pluralistic field was made by Buckton et al.[12], which integrated diverse insights via a cross-disciplinary 'regenerative lens'. The authors argued that key qualities required to sustain regenerative dynamics included an ecological worldview embodied in human action, mutualistic interactions between human and ecological phenomena, high diversity and agency, and reflexivity.

At present, literature on regenerative dynamics is scattered across many different fields in research and practice, and theoretically or empirically well-grounded concepts are sometimes mixed up with vague definitions, or ideological rather than academic reasoning. To address these shortcomings, this Review Article presents a formalization of regenerative dynamics and related concepts. With more and more papers referring to regenerative dynamics or other derivatives of that term (see Table 1 as well as ref. 12 for an additional, independently derived overview), it is important to build a body of terminology and theory that is precise and broadly applicable.

In this Review Article, we specifically do not take an ideological or political perspective on regeneration. Rather, the main contribution of this review is to lay out a general, internally coherent framework to study and manage regenerative systems, which can then be applied by scientists and practitioners across diverse fields. Using generally applicable, clear and consistent terminology, in turn, opens new ground for comparing regenerative systems across multiple scales and domains, as well as examining possible interactions among them. A generalized understanding of regenerative systems thus offers a new boundary object for sustainability science, that is, a robust but flexible concept that has the potential to engage different areas of science via collaboration towards shared goals[13].

The following sections define key terms and lay out examples of discourses from multiple domains on regenerative dynamics, showing how notions of regenerative dynamics and practices have begun to enter various domains. Building on this, a multi-domain framework of regenerative systems is introduced that spans scales from individuals to large-scale natural resource management systems.

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Table 1 | Examples of the wide range of overlapping terms currently used in different contexts to describe phenomena related to regenerative system dynamics

Term	Example usage
Regeneration of ecosystems	Regeneration can be understood as "the promotion of self-renewal capacity of natural systems with the aim of reactivating ecological processes damaged or over-exploited by human action"[21].
Regenerative agriculture	Regenerative agriculture (for example, agroforestry, permaculture) describes maintaining and improving resources through continuous renewal of the complex living system, having in mind that there is no 'waste' in nature[8].
Regenerative business	Regenerative businesses "enhance, and thrive through, the health of social-ecological systems in a co-evolutionary process"[6]
Regenerative capacity	Regenerative capacity is "the ability of nature to revive itself, recover from disturbances, and rebuild its functions"[21].
Regenerative compatibility	"Regenerative compatibility refers to the synergistic association between environmental attitudes and healthy ecosystems that triggers sustainability"[10].
Regenerative culture	"Regenerative culture is a culture that is consciously building the capacity of everybody in a particular place to respond and change"[57].
Regenerative design	Regenerative design is "a system of technologies and strategies, based on an understanding of the inner working of ecosystems that generates designs that regenerate socio-ecological wholes (that is, generate anew their inherent capacity for vitality, viability and evolution) rather than deplete their underlying life support systems and resource"[7].
Regenerative development	"A system of developmental technologies and strategies that works to enhance the ability of living beings to co-evolve, so that the planet continues to express its potential for diversity, complexity, and creativity through harmonizing human activities with the continuing evolution of life on our planet, even as we continue to develop our potential as humans"[7].
Regenerative economy	A regenerative economy is "an improvement of the entire living and economic model compared to previous business-as-usual economy and resource management"[79].
Regenerative organizing	Regenerative organizing is "the process of sensing and embracing surrounding living ecosystems, aligning organizational knowledge, decision-making, and actions to these systems' structures and dynamics and acting in conjunction, in a way that allows for ecosystems to regenerate, build resilience and sustain life"[53].
Regenerative sustainability	Regenerative sustainability aims for thriving living systems from the scale of individuals to the entire Earth system[80].
Regenerative interventions for ecosystem restoration	Management interventions (such as introducing target species, readjusting abiotic conditions) aim at (re-)instating ecosystem functions and services as well as particular suites of species[23,81].

The table shows that somewhat similar terms are used in slightly different ways in different contexts. Building loosely on the terms outlined here—but not adopting them directly—this Review Article seeks to provide a consistent set of definitions that can be used across many different sustainability-relevant contexts.

Terminology and key concepts

Consistent terminology is vital to move ideas around regenerative dynamics from fringe discourses to the mainstream of sustainability science. For this reason, the terminology proposed here is internally consistent and broadly transferable. This terminology loosely builds on a range of terms from diverse domains, where similar ideas have been described, with varying levels of precision, using diverse variants of similar wording (Table 1).

The basics of regenerative dynamics

'Degeneration' can be defined as the decline of a desired outcome variable. It is associated with numerous sustainability problems in contexts of both the environment and human well-being; widespread degeneration has been observed across different contexts, for example, in ecosystems[14], mental health (that is, climate anxiety[15]), community cohesion[16], food security[17] and agricultural soil quality[18]. By contrast, regeneration can be understood as the process of renewal or re-creation of a certain desired outcome (noting that no outcome can ever be recreated precisely). For example, regeneration describes the renewal of strength and vitality of a patient after sickness or the recovery of an ecosystem after a heavy disturbance.

Regenerative dynamics, in turn, occur when desired outcomes, such as social well-being or soil health, regenerate in a system not only once, but over and over[12]. Regenerative dynamics thus can be thought of as an upward helix for a specific context (Fig. 1); a system with regenerative dynamics in a certain outcome will produce increasingly higher levels of that outcome over time or at some point maintain a steadily high level of the outcome. Regenerative dynamics are maintained partly by momentum within the system. As such, regenerative dynamics are partly self-perpetuating and endogenous but are shaped and constrained by contextual and interconnected conditions and require resources such as energy, labour or materials to be sustained. A simple example of regenerative dynamics is that of an ecosystem after a major disturbance. A mature forest that was hit by a wildfire, for example, will undergo partly self-perpetuating regenerative dynamics until structural complexity and species diversity are once again high many years later[19]. For this, it requires energy input from the sun, as well as appropriate nutrients, precipitation and a source of seeds. Contextual conditions (such as climate and weather conditions in this case) thus influence the speed, duration, direction and nature of regenerative dynamics; in the case of a regenerating forest, for example, periods of drought will favour a different set of species than will periods of wet weather.

Importantly, in a context of sustainability, regenerative dynamics pertaining to one particular desirable outcome should not be premised on the ongoing degeneration of other desirable outcomes. For example, an upward helix of economic activity (economic growth) cannot be considered regenerative from a sustainability perspective if it is premised on the depletion of natural capital stocks. Depletion can occur because a non-renewable natural stock is being exhausted or because a renewable natural stock is used at a rate that is faster than its capacity for renewal. A regenerative system thus is defined by exhibiting internal regenerative dynamics without negative spill-over effects on other systems and their dynamics[12]. Moreover, a regenerative system in principle can be maintained (given sufficient energy input; see the preceding paragraph) in the long run; that is, it does not undermine its own long-term potential for ongoing regeneration by depleting the resources on which it depends.

Ultimately, regenerative dynamics face natural limits when a certain outcome cannot further increase. For example, in the context of agriculture, natural soil fertility can be managed to increase but will eventually be limited by mechanisms and rates of weathering and rates of biological activity within the soil[20]. At such natural limits, regenerative processes will not cause ongoing increases in the outcome variable.

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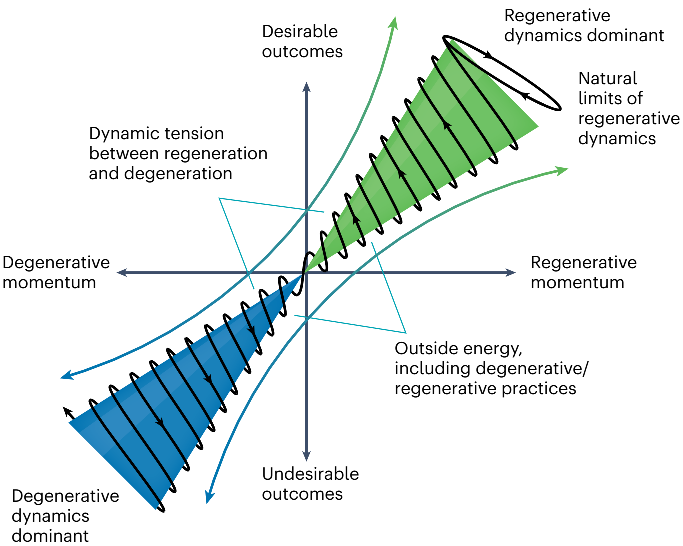

Fig. 1 | Generalized view of regenerative dynamics. The stylized depiction shows a partly self-sustaining system with a certain level of inertia. Outside energy or practices are needed to maintain regenerative momentum. Through time, regenerative dynamics will lead to the increase in an outcome variable until natural limits are reached. Figure adapted with permission from ref. 22, Taylor & Francis.

Regeneration in relation to restoration

Regeneration differs from restoration[21]. Linguistically, restoration comes from the Latin verb 'restaurare', which means to repair, give back or build up again; regeneration comes from 'generare', which means to give birth or generate[21]. 'Restoration' thus refers to the conscious but exogenous process of recreating a certain state, or repairing part of a system, whereas in regeneration this process is primarily endogenously driven. For example, whereas regeneration suggests humans co-evolving with and participating as nature, restoration refers to humans doing things to nature[22]. Remediation is a type of restoration; an area can be remediated after pollution (for example, from mining operations) and then depolluted, such that contaminants or toxic compounds are present below deleterious quantities and some form of vegetation cover is reinstated. Repair, furthermore, is a technical view and one step of a possible remediation. A broken leg, for example, can be repaired by a surgeon, but regaining the functionality to walk with the leg (remediation) will require more than an external fix and hinges on internal regenerative dynamics. Repair and remediation, in such cases, might be a prerequisite for regenerative dynamics to unfold but may not achieve full restoration to the former state. As such, we do not consider the action of restoration itself to be regenerative unless the system resulting from restoration is, at least to an extent, dynamically self-perpetuating. In degraded forest ecosystems, for example, tree planting can be a restoration activity, but it is regenerative only if the trees planted can naturally regenerate thereafter or foster other types of regenerative dynamics.

With repair, remediation and restoration being possible prerequisites for regenerative dynamics to unfold, different management interventions thus span a broad restorative continuum, ranging from activities with relatively low levels of ambition that are only remedial in nature to more ambitious activities that seek to support self-sustaining, regenerative dynamics[23]. Such a continuum is widely recognized in restoration ecology[23], but similar thinking has also entered the management arena. Here, the R-strategies 'reduce', 'reuse' and 'recycle' are relatively less ambitious ways to lower a firm's negative footprint on the environment. On the more ambitious end of the spectrum, the terms 'restore', 'regenerate' and 'rewild' are used to help shift the management focus towards creating a net-positive handprint[24], that is, the sum of positive actions that a firm takes to improve life on the planet.

Regenerative practices

Some systems exhibit regenerative dynamics naturally (for example, ecosystems after disturbance or animals that regularly sleep). In many anthropogenic systems, however, human actions are required to initiate, sustain or strengthen regenerative dynamics[12]. To account for the critical role of human agency[12], the term 'regenerative practices' is useful to denote intentional activities that support regenerative dynamics. Regenerative practices constitute, first of all, a type of effort, energy or material input into a given system. Regenerative practices typically are not one-off actions but require sustained efforts to support certain desired system dynamics. For example, renewable energy infrastructure requires initial capital investment, which can then generate profits that in turn allow for more renewable energy investment and cost reductions through economies of scale[25]; regenerative agriculture requires a farmer to regularly use knowledge, energy and resources (for example, organic fertilizers) according to agroecological principles when growing and harvesting crops[8,26]; many of the benefits of meditation also require ongoing practice to reap the benefits[27,28]. The definition of regenerative practices thus is consistent with the sociological definition of practices—that is, "sets of hierarchically organized doings/sayings, tasks and projects" (ref. 29, p. 30)—and implies four key elements: (1) a practical understanding of how the regenerative dynamics of something work, as is emerging in different academic domains (compare Table 1); (2) rules or guiding principles for how to perform regenerative actions; (3) a teleo-affective structure clarifying what is and what is not desirable; and (4) a general understanding about what regeneration and regenerative practices are, as developed in this paper.

Notably, not all systems will be equally responsive to regenerative practices. Regenerative momentum is the existing inertia of the system in the direction of regeneration, whereas degenerative momentum is the inertia of the system in the direction of degeneration. Strong momentum thus means a strong tendency to continue with the existing dynamics. Depending on the existing dynamics, regenerative practices therefore need to put a lot of time, effort or other resources into an existing system to 'turn it around'—from a downward helix towards an upward helix (Fig. 1). Severely degraded soils thus cannot be turned into fertile living systems overnight and may never reach their previous state[31]; violent situations require active efforts at (for example) non-violent communication to turn around into regenerative dynamics[32]; and environmentally harmful economic activities may be upheld by institutions that can be highly resistant to change[33]. In fact, systems that are particularly degraded can be severely resistant to change because they are stabilized in their degraded state through certain feedback mechanisms[34,35]. In such cases, undesired feedbacks first need to be interrupted and then be replaced by desired, regenerative dynamics[36].

Combining the different concepts defined in the preceding, an archetypical regenerative system can be understood as a (1) partly self-sustaining system with strong endogenous dynamics and a certain level of inertia but which (2) draws on outside energy or resources to maintain its momentum. Energy or resource input may be non-anthropogenic (for example, sunlight) or (3) anthropogenic, drawing on intentional regenerative practices. Through time, regenerative dynamics will lead to (4) the increase in an outcome variable, until (5) limiting dynamics switch from outcome growth to outcome maintenance (Fig. 1).

Examples and applications

On the basis of the preceding definitions, regenerative dynamics, and in many cases associated practices, have been observed or theorized in a range of sustainability-relevant domains (Table 1; see also refs. 12,37,38). In the following, we discuss how regenerative dynamics and practices are understood or theorized in six domains (agriculture, psychology, economics, management, sociology, social–ecological systems) where the notion has recently emerged. We acknowledge that there are other domains (such as medicine, development, urban design and education) where regenerative dynamics and practices are also discussed and which also have notable parallels[12].

Some of the strongest evidence for regenerative dynamics exists in the domain of agriculture[8]. Regenerative agriculture arose as a counterpoint to conventional or synthetic chemically led agriculture. According to Rhodes[39], regenerative agriculture has "at its core the intention to improve the health of soil or to restore highly degraded soil, which symbiotically enhances the quality of soil, water, vegetation and land-productivity". Regenerative agriculture thus seeks to improve life and biological processes in soils (the specific outcome the practice focuses on), including biodiversity, carbon content and ability to filter and hold water. Specific actions include maximizing the time crops spend covering the soil, increasing crop diversity in space (intercropping) and time (diverse rotations) and minimizing soil disturbance. Additional actions are cover-cropping, limited or no tillage cropping, agroforestry and rotational grazing, depending on topography, soil type and other local conditions[40]. Importantly, regenerative agricultural practices have been inspired by many philosophies, principles and knowledge from diverse people on all continents (for example, countries in South America, Africa and Asia), including from Indigenous peoples. For example, as described by a study conducted in Brazil, considering the Kawaiwete (Tupi–Guarani) and the Ikpeng (Carib–Arara) peoples in the Xingu Indigenous Territory (state of Mato Grosso)[41], some local practices based on indigenous knowledge demonstrated great potential to facilitate natural regeneration—for example, selecting appropriate areas for agricultural use, limiting the number of cultivation cycles, protecting and selecting individual trees during the cultivation period and attracting seed dispersers.

Another domain that has started to investigate regenerative dynamics and practices is psychology, with growing interest in the practice of mindfulness. Mindfulness refers to a diverse set of individual practices to counteract the effects of stress on individuals, to improve their well-being and to connect with others and with the environment[27]. Key is the purposeful focus on the present moment and the non-judgemental nature of the observation of inner sensations, emotions and thoughts. Practising mindfulness supports individuals in becoming aware of and understanding their patterns of thoughts and emotions. Observing their wandering mind during meditation also helps people to achieve better control over their attention. The best-known and researched clinical programmes are called mindfulness-based self-regulation (MBSR) and mindfulness-based cognitive therapy (MBCT). Research suggests that MBSR helps people in their self-regulation, specifically in their control over their attention and repetitive thoughts[42]. MBSR has been influential in the self-regulation of stress, anxiety, chronic pain and various illnesses[43]. A recent meta-analysis over 23 randomized controlled trials showed that MBCT and MBSR led to short-term improvements of clinical anxiety when compared with treatment as usual[44]. Although the term 'regenerative' is rarely used in academic work, the notion of an 'upward spiral' is used by some mindfulness researchers to describe the dynamic improvements in well-being and other resources[45] that may counteract the stifling effects of climate anxiety, for example. Mindfulness could ultimately influence not only the well-being of individuals but also that of groups (Box 1).

Ideas of regenerative dynamics and practices are also rapidly increasing in the domain of economics. Here the notion of a regenerative economy is based on at least two different perspectives. The first understands the economy as a metabolizing subsystem of the environment[46] and prioritizes the need to move away from environmentally damaging linear throughput of energy and material continually extracted from the environment, drawn into the economy and then returned to the environment as waste. In this perspective, typified by the circular economics approach, regenerative economics is premised on building systems that (re)generate value flows within the economy by retaining energy and materials within the system[47]. Regenerative practices, in this perspective, are associated with the design of more sustainable products, or more sustainable supply chains, reuse, remanufacturing or recycling processes[48]. Notwithstanding these developments, the existence of fully self-sustaining practices—as opposed to simply more metabolically efficient practices—is ultimately questionable[49] (Box 2). A second perspective frames regenerative economics as a form of biophilic design[50], which encourages diverse practices that

BOX 1

Scaling up regenerative practices: mindfulness for individuals and groups

Mindfulness seems to benefit individuals in preventive settings. In its 'broaden-and-build theory of positive emotions', Fredrickson and Joiner[45] proposed an upward spiral through which individuals can increase their psychological, social, cognitive and health resources. The starting point of the upward spiral is the experience of positive emotions that are brought about by practising gratefulness or loving kindness meditation on a regular basis[45]. In a field study with working adults, Fredrickson et al.[27] showed that daily practice of loving kindness meditation contributed to an increase in participants' mindfulness, purpose in life and social support. In turn, these increments in personal resources predicted increased life satisfaction and reduced depressive symptoms. A more recent meta-analysis over 56 randomized controlled trials suggests that mindfulness-based interventions at work effectively improved employees' mindfulness, well-being, compassion and job satisfaction. Results were maintained in follow-up assessments several months later[82].

Mindfulness can be upscaled by training teachers[83], who introduce practices and programmes of mindfulness into classrooms, which can benefit both students and teachers. In the Breathing Break Intervention for primary school children, for example, teachers were trained in mindfulness and asked to invite their students to perform breath-based mindfulness exercises of 3–5 minutes each day. Teachers could choose between activating and calming practices depending on the situation in the classroom. Evaluation results of this randomized controlled trial indicate that these breathing practices contributed to the improvement of girls' prosocial behaviour and the maintenance of a positive classroom climate during the COVID-19 pandemic, when class cohesion was challenged by quarantines and social distancing rules. Because nearly all teachers and students enjoyed the Breathing Breaks, 53% of the children wanted to continue them in class after the intervention. While schools were closed during a lockdown, a sizable minority of 18% of the children in the intervention group shared the breathing exercises with family members and continued to do them on their own more than 'once a week'[84]. These results speak for the self-perpetuating nature of these breathing exercises, although long-term effects need to be confirmed by further research. Professional development can stimulate teachers and other employees to change their attitudes along with organizational procedures of schools and workplaces[85]. Teaching leaders in organizations to use contemplative practices to work on their nature connectedness and their inner development goals shows promising first results[86].

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BOX 2

The promises and limits of material regeneration in a circular economy

Circular economies seek to recirculate materials within the economic system, either by reusing the products themselves or by 'recovering' the component parts of those products for reuse in new products. This approach is inspired by the (seemingly) zero-waste cyclical flows of energy and materials in natural ecosystems. The assumption is that recirculation can result in the identical amount, or quality, of economic products without drawing more material and energy into the economy. However, there can be no perfectly circular flow of resources, materials and products because of the laws of thermodynamics. Waste (increased entropy) is an inevitable by-product of any work done on the materials used within the economy (from extraction, through manufacturing, to their usage and eventual recycling), leading to a decrease in the quantity and/or quality of those materials[87]. For chemical elements (such as metals or phosphorous), regeneration is not possible because these resources are generated within the life cycles of stars. For some biological materials such as cellulose or cotton, regeneration of harvested resources is possible but still requires external inputs (for example, water, fertilizers, energy). The more matter we mobilize and circulate within the economy, the more energy and materials we will need to maintain that circulation and the faster the degradation of that matter will occur, resulting in yet more waste and the need for yet more energy. There is much that can be done to minimize material waste and maximize regeneration; however, there are also inevitable biological, chemical and physical limitations to the regeneration of materials in our closed ecological–economic system.

mirror natural systems and thereby facilitates the building of natural capital assets via natural processes to ultimately create an 'economy in service to life'[51]. One example of biophilic design (also referred to as nature-based solutions) is the (re)building of wetlands for water purification/regulation, with co-benefits in terms of carbon sequestration, biodiversity conservation and nutrient cycling[52].

In the management domain, the notion of regeneration is used to rethink how firms do business. Here regenerative dynamics capture the re-embedding of organizations in their social–ecological context to create new forms of value[53]. Regenerative businesses break with linear thinking and human–nature dualism and, in turn, are socially and ecologically embedded in a 'place'[54]. A central premise is that regenerative businesses enhance, and thrive through, the health of social–ecological systems in a co-evolutionary process[6,12]. Such net-positive thinking requires managers to place societal and environmental well-being above profits and to assess how and where innovations are able to support sufficiency instead of growth[55]. Regenerative organizing thus needs to deal with all the place-based paradoxes of being embedded in a social–ecological system[53,54]. This suggests synchronizing business practices across space and time with nature instead of seeking to control nature. The 'restore–preserve–enhance' scale offers examples of such regenerative business strategies trying to enhance social–ecological well-being while making a profit[6]. Regeneration as part of the R-strategies[56] offers managers a heuristic to deeply rethink sustainability in management beyond its closest stakeholders and with a stronger focus on planetary health and societal well-being.

Within the sociology domain there is a focus on regenerative culture as a means of consciously building the capacity of everybody living in a particular place to engage and participate[57]. Regenerative practices thus describe self-organization that leads to thriving communities[58]. Regenerative design refers to a more participatory, biophilic, sometimes even biomimetic, perspective to design life-giving patterns, healthy processes and actions in a living system in a particular place[22]. Furthermore, regenerative governance includes holacratic and sociocratic models of organizing that emphasize self-management and distributed authority and power, thereby expanding the adoption of living system principles to the management of larger systems (for a more elaborated overview, see ref. 58).

Finally, within the domain of social–ecological systems, the notion of a regenerative society[59] seeks to fundamentally rethink and redefine human–nature relationships. In this context, the regenerative dynamic intends to support transformation practices by promoting mutualistic social–ecological relationships[10,60,61] and by re-negotiating what mutualistic and reciprocal mean[62]. Research on social–ecological systems uses notions of regeneration as a core framing to make explicit the co-dependence of people and nature[61], emphasizing the synergies among healthy, diverse ecosystems, environmental attitudes and the restorative benefits of nature on people's quality of life[10,12].

As highlighted in the preceding, although perspectives on regenerative dynamics and practices vary across domains, and terminology can differ substantially (Table 1), there are striking similarities in how scholars have conceptualized regenerative dynamics in different domains[12,37]. Three core similarities are (1) the intention to positively enhance the state of a certain desired outcome variable (2) through systems-focused (and often place-based) regenerative practices, which (3) build on human–nature connectedness. The convergent interest in regenerative dynamics across fields is especially remarkable because the way regenerative practices are investigated methodologically is highly diverse, encompassing qualitative, quantitative and mixed-methods approaches. Still, all of the domains outlined in the preceding essentially hypothesize a dynamic similar to that depicted in Fig. 1. With these similarities, an obvious question arises: can regenerative dynamics be linked across domains to boost sustainability initiatives?

Cross-domain links of regenerative dynamics

The overview presented in the preceding shows that regenerative framings have begun to be used in diverse sustainability domains. These domains differ in their focus, but also in the scale at which they operate. For example, the psychology domain applies most immediately to individual persons, although it also can be scaled up via coordinated interventions targeting schools, firms or other groups of people (Box 1). By contrast, regenerative management relates largely to the scale of individual firms and their area of influence; regenerative agriculture or ecosystem restoration can take place within single plots or across entire landscapes, thus operating at various system scales depending on the context; while the notion of a regenerative economy could, in principle, guide economic decision-making at the level of nation states and beyond (Fig. 2).

The diversity of domains and scales spanned by regenerative framings means that a focus on regenerative practices could be a powerful new boundary object for sustainability at large. Multiple existing frameworks highlight the vital importance of multi-domain, cross-scale interactions for sustainability transformations. For example, the socio-technical transitions framework by Geels and Schot[63] considers that technical innovations emerge in societal niches, which shape higher-level socio-technical regimes. These regimes, in turn, are fundamentally constrained by the even larger-scale socio-technical landscape, which, in turn, continuously evolves out of different co-existing regimes. Similarly, the panarchy model by Gunderson and Holling[64] considers nested cyclical dynamics within ecological systems—again, noting that slower, larger-scale cycles fundamentally stabilize and

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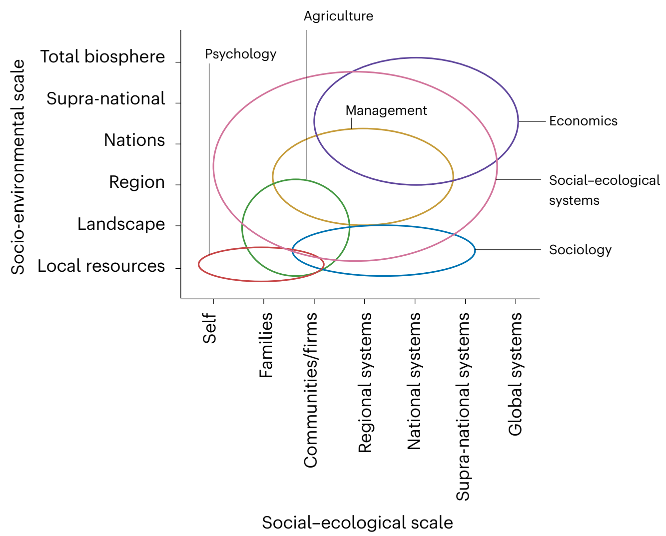

Fig. 2 | Regenerative practices are potentially linked across domains and may positively influence one another. Change may emerge from the bottom up. For example, an individual regularly practising mindfulness may make different dietary choices, with implications for agricultural systems. Changed agricultural systems, in turn, could influence higher-level social–ecological interactions, including improved biodiversity outcomes. Change can also emerge from the top down. For example, changes in large-scale economic systems could incentivize individual firms to act in more regenerative ways or incentivize more regenerative agricultural practices. Regenerative 'ripple effects' thus can permeate up and down social and biophysical scales. Figure adapted with permission from ref. 65, Elsevier.

constrain faster, smaller-scale cycles, while faster, smaller-scale cycles are the primary origins of disruption and innovation. Both of these models are hierarchical and describe nested dynamics in that they see the primary impetus of change originating from smaller scales while larger scales provide stability.

Not all conceptual models of sustainability transformation are strictly hierarchical. Everard et al.[65], for example, considered that ripple effects could spread upwards as well as downwards across interlinked social and environmental scales. Thus, although hierarchical relationships are likely to be important within a world of ever-present tension between structure and agency, regenerative practices might offer a simple but useful starting point for future research on cross-domain regenerative dynamics, bi-directional ripple effects and healthier human–nature interactions. Research shows, for example, that individuals who regularly engage in mindfulness practices act differently with respect to sustainability within their communities than individuals who do not (for example, through different dietary choices)[66,67], and it seems plausible that landscapes dominated by regenerative agriculture and ecosystem management go hand in hand with the mindsets held by individual people living there[68,69], in different ways than in industrialized farming landscapes with degenerating human–nature connectedness[70]. Hence, there could also be a key role for regenerative businesses, such as community-supported agriculture organizations, to lead human and ecological regeneration for a regenerative economy to emerge. A central hypothesis thus emerging from this paper is that regenerative practices across domains might mutually reinforce one another, favouring increasingly sustainable dynamics across multiple scales and domains (Fig. 2).

Outlook and emerging questions

To mainstream thinking about regeneration and to develop a heuristic that is useful across multiple academic domains, regenerative systems research should draw attention to the interplay of regenerative and degenerative practices, momentum and dynamics across multiple spatiotemporal scales. That is, regenerative dynamics cannot be studied in isolation; rather, such dynamics arise from the interplay of regenerative versus degenerative practices, within a context of regenerative or degenerative momentum, embedded within other complex dynamics of social–ecological systems. Buckton et al.[12] recently equipped researchers with a set of principles when applying a regenerative lens. This paper further underlines the role of regeneration as a boundary object and offers a dynamic framework to inform new multi-domain, cross-scale research, with an emphasis on how to actively foster desirable changes in outcome variables over time.

The framing of sustainability challenges around regenerative practices offers a constructive, encouraging and hopeful way to think about and find solutions to the many interconnected grand challenges that humanity now faces[1]. Many established ways of thinking about sustainability, including the Sustainable Development Goals (Fig. 3), are somewhat static in the way they approach the definition of goals. Instead of focusing on static goals, a focus on regenerative practices emphasizes the need for positive dynamics in multiple interconnected domains (thereby supporting 'nature positive' or 'climate positive' aspirations)[37]. Behaviourally oriented psychologists emphasize that human beings (and other organisms) are set up in a way that fulfilling needs and reaching positive (sub-)goals positively reinforces a given behaviour, which means that it will occur more frequently in the future. Positive reinforcement (and feedback) stimulates proactive behaviours and creativity for further improvements whereas avoiding a future catastrophe is a negative form of reinforcement, which is much less motivating[71]. Consistent with this, systemically oriented psychologists conceptualize positive emotions generated by reaching a goal to be part of an upward spiral[45]. According to positive psychology, any combination of positive experiences or emotions, engagement, relationships, meaning and purpose, and accomplishments contribute to a pleasant, happy life[72]. The framing offered in this paper thus could be helpful to engender new research, creativity and agency towards increasingly better sustainability outcomes, including for key themes captured by the Sustainable Development Goals (Fig. 3).

An integrated focus on regenerative dynamics and practices could not only lead to new insights about complex systems' behaviour, but also foster the evolution of new types of goals, mindsets and ways of interacting with nature—which are recognized as potentially deep leverage points for a transformation towards sustainability[73]. For example, regenerative co-design of human–nature interactions could be a new cross-domain goal for sustainability science and practice. With a focus on regeneration, mindsets may change from a focus on maintaining some kind of balance between humans and non-human nature towards healthy, positive relationships among many kinds of different entities in diverse systems[12]. Arguably, such a shift in mindsets has already begun, as shown by the rapidly growing body of work on sustainability-relevant relationships, for example, in the context of the Nature Futures Framework of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services[30,74] or in the context of the One Health framework, which emphasizes relationships among humans, (other) animals and the environment[75]. Similarly, recent management and entrepreneurship studies emphasize the need to more explicitly consider interlinkages between organizations, markets, society and the natural environment[53]. Consistent ideas are emerging even in chemistry and its industrial application; for example, sustainable chemistry is generating a new understanding of inclusion of all stakeholders along the full life cycle of products[76,77].

In conclusion, the growing focus on regenerative systems is a logical continuation of existing framings and shifts that are under way to better address a multitude of complex and interrelated sustainability problems. While the term 'sustainability' will undoubtedly remain dominant for the time being, the concept of regenerative systems

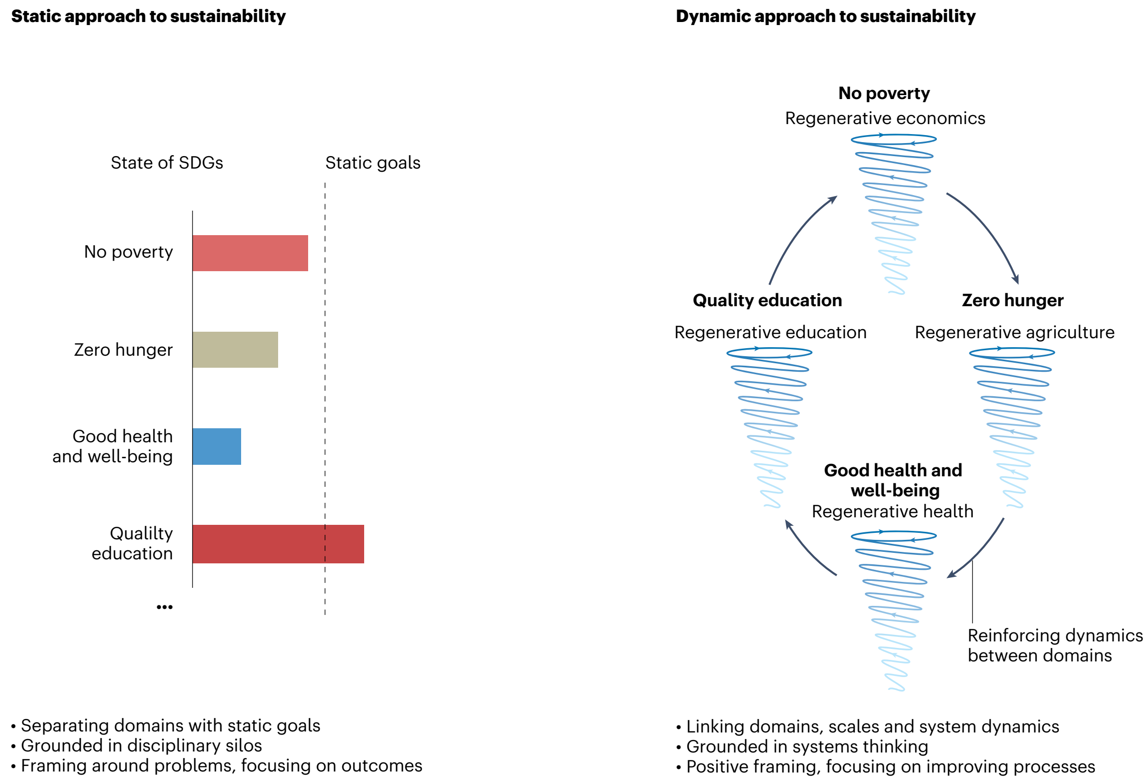

Fig. 3 | Summary of key features and possible contributions of a focus on regenerative dynamics and practices for sustainability. A focus on regenerative dynamics and practices could help to move the focus from attaining static goals towards emphasis of ongoing, dynamic improvements across multiple interacting domains. Note that the domains shown here are only hypothetical examples; specific, actual links across domains would need to be established through new research. SDG, Sustainable Development Goal.

promises to be an increasingly important element and complement[22,58]. A regenerative systems focus is an explicit way to emphasize that greater efforts are needed to foster partly self-perpetuating dynamics that are life-affirming at all scales and that support human well-being.

Numerous questions for future research arise in response to the ideas put forward here. For example, how prevalent are cross-scale and cross-domain interactions of regenerative dynamics? Contrary to the preceding discussion, could interactions across domains or scales hamper rather than positively reinforce one another, thus leading to unintended, perverse outcomes? Are small-scale regenerative dynamics nested within larger-scale regenerative dynamics ('spirals within spirals, composed of spirals'), or what is the nature of scale relationships among different types of regenerative systems? How do regenerative systems evolve or otherwise develop through time? Can key insights from one domain or scale be readily transferred to other scales or domains? Which methods are most suitable to the study of regenerative dynamics, and how can they be applied to specific problems (for example, sustainable food[78] or transport systems, rangeland management or chemistry)?

We started this paper by highlighting that interest in regenerative dynamics is growing in multiple, diverse domains. With the generalized framework of regenerative systems proposed here, and complementary to other recent advances proposed independently[12,37], we seek to inspire new thinking, research and action so that widespread nascent interest in regeneration can grow and be linked across currently disparate fields—hopefully giving a much-needed boost to sustainability research and practice.

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J.F., S.F., V.Z.Z. and D.J.A. led the conceptualization and writing of the paper. M.v.S., S.S., B.M.-L., V.M.T. and K.K. contributed text based on their disciplinary expertise and edited the manuscript.

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The authors declare no competing interests.

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